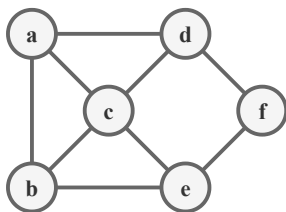
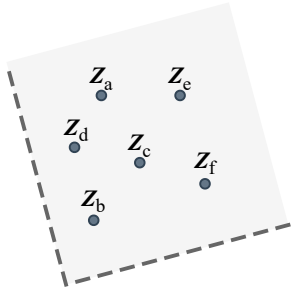
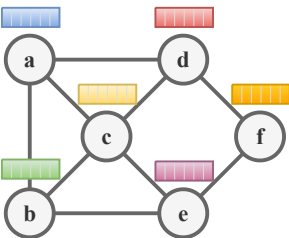
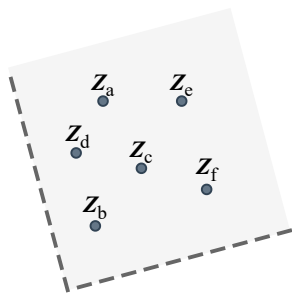


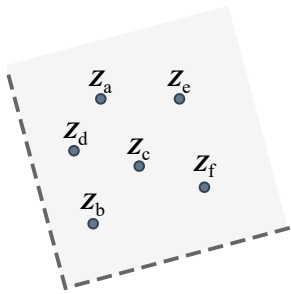
Graph
embedding

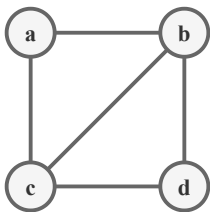


Shallow
embeddings

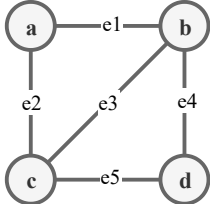


GNNs





	a	b	c	d
a	0	1	1	0
b	1	0	1	1
c	1	1	0	1
d	0	1	1	0

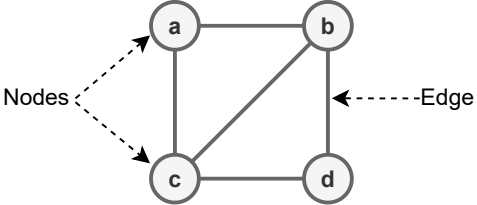


	e1	e2	e3	e4	e5
a	1	1	0	0	0
b	1	0	1	1	0
c	0	1	1	0	1
d	0	0	0	1	1

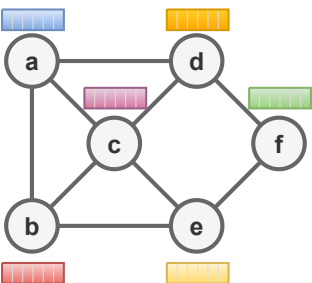
a	b	c	
b	a	c	d
c	a	b	d
d	b	c	

a	b
a	c
b	c
b	d
c	d

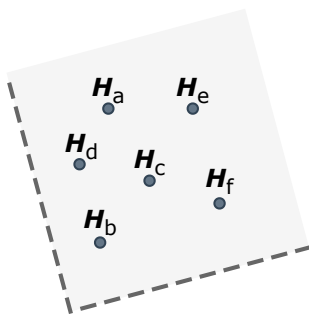
a	b
a	c
b	c
b	d
c	d



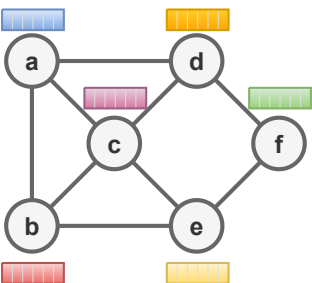
Graph Neural
Network



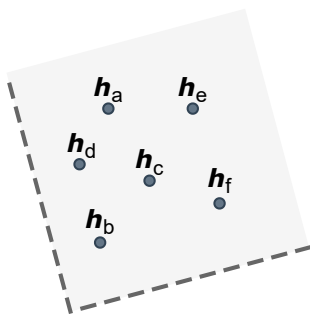
GNNs



$$\mathbf{H} = \text{GNN}(\mathbf{X}, \mathbf{A})$$



GCN



$$\mathbf{H} = \text{GNN}(\mathbf{X}, \mathbf{A})$$

Techniques	Dataset	Applications	Evaluation Metrics
Combined DeepWalk, LINE, Node2vec, and SDNE [26]	MATADOR, PubTator, and BioGRID	Link prediction	AUC, AUPR, MAP, Avg. R-precision, and Precision@k
HeteWalk [115]	HPRD, MISIM, MimMiner, Dis-GeNET, and miRTarBase	Link prediction	AUC
Cascade model [97]	BioChem, Drug Bank, and PubChem	Link prediction	Accuracy, hits@10, and AUC
[29]	Krogan, Dip, and BioGRID	Node clustering	Precision, recall, F-score, fraction, geometry accuracy, and MMR
HNERMDA [102]	MDAD and aBiofilm	Link prediction	Accuracy, AUC, and AUPR
PmDNE [30]	HMDD3.0	Link prediction	AUC, AUPR, precision, accuracy, recall, and F1-score
HO-VGAE [27]	HI-II-14, HI-III, Lit-BM-13, BioGRID, and Bioplex	Link prediction	AUPR, Precision@k
HMNE [105]	Lazega, CKM, DBLP, C.elegans, H.genetic, PPI, and Twitter	Link prediction and node classification	F1-micro, F1-macro, and AUC
TriModel [99]	DrugBank_FDA, KEGG_MED, and Yamanishi_08	Link prediction	AUC and AUPR
FactorHNE [103]	DisGeNet, HPO and Orphanet, STRING 10	Link prediction	AUPR, AUC, Precision@K, and Recall@K
[100]	DrugBank_FDA, UNIPROT	Link prediction, node clustering	Accuracy
Hybrid model GVS [31]	GO, HPRD, CTD, HMDD and MATADOR	Link prediction	Accuracy and F1-score
DeepWalk and Node2vec [98]	DrugBank, Bio2RDF, human disease network, SIDER, KEGG, and PharmGKB	Link prediction	AUC and AUPR
Netpro2vec [92]	LFR, MREG, Kidney RNASeq, Brain fMRI COBRE, Breast RNAseq, Breast Microarray, MUTAG	Node classification	Accuracy, precision, recall, F-score, and MCC
BiLSTM [101]	Human, DUD-E, and ChEMBL	Node classification	AUC, precision, and recall
scLINE [104]	Usoskin, Li, Pollen, Patel, Darmanis, Camp, Muraro, and Petropoulos	Node clustering	DBI, NMI, ARI, Jaccard and Purity
PRD [34]	Bio2RDF, and DDI Corpus	Link prediction	AUC, AUPR
ACNE and ACNE-ST [116]	Cora, Cite-seer, Wiki, and DBLP_C4	Node classification and node clustering	F1-micro and F1-macro
Pathway2vec [33]	EcoCyc, Human-Cyc, AraCyc, YeastCyc, LeishCyc, and TrypanoCyc	Link prediction and node clustering	F1-micro
LPPI [28]	PPI network and GraphSAGE-PPI	Link prediction and Node Classification	Accuracy, sensitivity, precision, MCC, and AUC
MRMTI [32]	miRTarBase, miR-Base, HumanNet, and biomaRt	Link prediction	AUC, AUPR, precision, recall, F1-score, and balanced accuracy
CANE [117]	Disease Encyclopedia Section of XYWY.com.	Link prediction	Precision@k and recall@k